Thermodynamic description of the Ag–Ni system: a synthetic benchmark document

ORCHID synthetic benchmark (generated; no human authors)

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Abstract

This synthetic document describes the Ag–Ni binary system as calculated with pycalphad from the SGTE dataset AgNi-04Din.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key. The highest invariant reaction, L2 → L1 + (Ni), lies at 1703.4 K.

1. Introduction

The Ag–Ni system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as miscibility gap, monotectic. The thermodynamic description is the SGTE collection entry (Hallstedt, 2025), which reproduces the assessment of A.T. Dinsdale, unpublished, 2004. The calculations use pycalphad 0.11.2 at 1 atm.

The melting point of Ag reproduced by the dataset is 1234.9 K. For Ni, the liquidus meets the axis at 1728.2 K.

2. Phase diagram

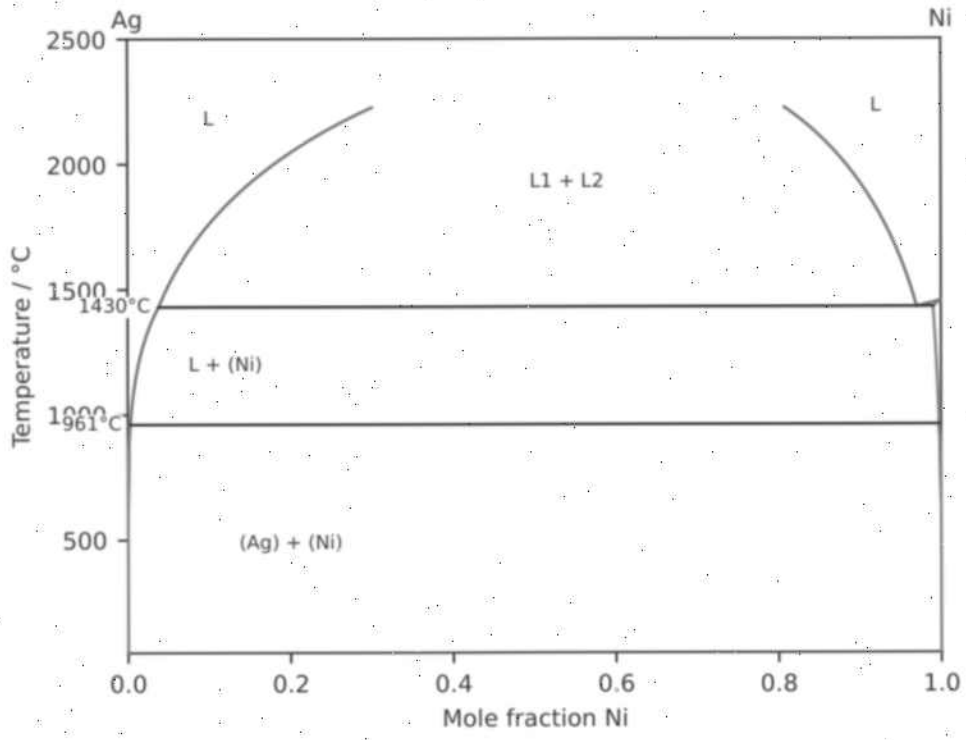


Fig. 1. Calculated Ag–Ni phase diagram.

The dataset places a monotectic (L2 → L1 + (Ni)) at 1703.4 K; the composition of L1 is 3.8 at.%. The dataset places a eutectic (L → (Ag) + (Ni)) at 1234.2 K; the composition of L is 0.3 at.%.

Table 1. Invariant reactions in the Ag–Ni system (calculated).

Reaction	Type	T / K	Compositions (x(Ni))		
			phase 1	phase 2	phase 3
L2 → L1 + (Ni)	monotectic	1703.4	0.970	0.038	0.990
L → (Ag) + (Ni)	eutectic	1234.2	0.003	0.003	0.997

3. Thermodynamic properties

At 2600 K the integral mixing enthalpy of the liquid has its maximum, 15118 J/mol, at 38.0 at.% Ni.

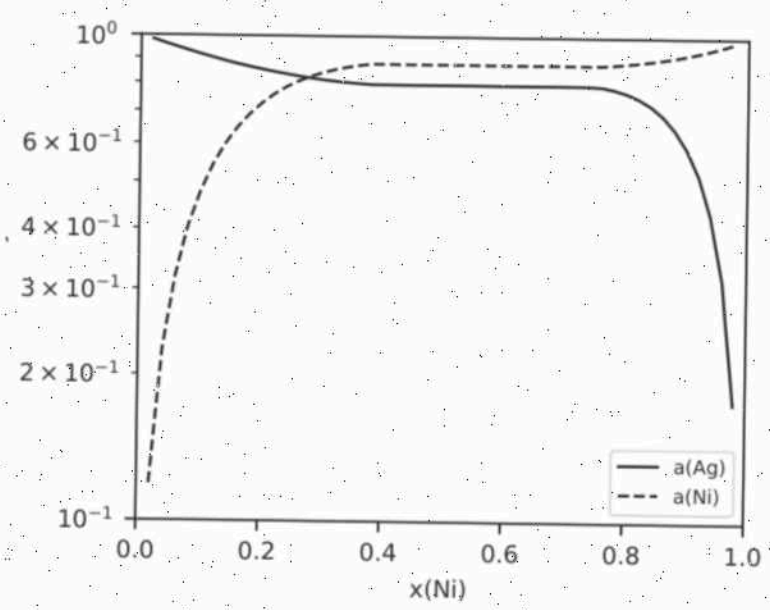


Fig. 2. Calculated activities in the liquid at 2600 K (reference: pure liquids).

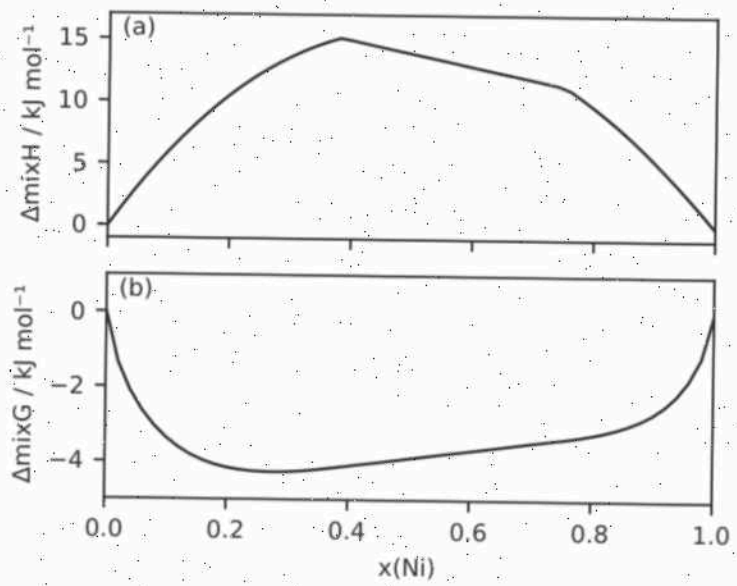


Fig. 3. Calculated (a) mixing enthalpy and (b) mixing Gibbs energy of the liquid at 2600 K.

Table 2. Redlich–Kister parameters of the liquid (Ag–Ni).

	Liquid: L(ν) = a + b·T + c·T·ln T		
ν	**a**	**b**	**c**
L0	63052.65	−7.119265	0.0
L1	4654.55	−3.636835	0.0

4. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.